

4. (a) **Choose words from the list below** to complete the following sentences.

ATP cytoplasm DNA nucleus protein vacuole

(i) Chromosomes are found in a cell structure called the [1]

(ii) Genes are sections of a long molecule with a double helix shape called [1]

(b) A grower crosses two plants, each with pink flowers.



The number of offspring plants is shown in **Table 4.1**. Some of the offspring had white flowers.

Table 4.1

Flower colour	Number of offspring
pink	21
white	7

(i) Calculate the ratio of pink flowers to white flowers. [1]

ratio of pink flowers to white flowers = :



Examiner
only

(ii) Complete the **two** sentences below by underlining the correct term for each.

I. The results of the cross show that the genotype of the parents is: [1]

- homozygous dominant
- homozygous recessive
- heterozygous

II. The genotype of the offspring with white flowers is: [1]

- homozygous dominant
- homozygous recessive
- heterozygous

(c) (i) In humans, the X and Y chromosomes are known as the sex chromosomes. Complete **Table 4.2** by placing **one** tick (✓) in each row to show the sex chromosome that may be present in the gametes. [2]

Table 4.2

Sex	Sex chromosome that may be present in the gametes		
	X only	Y only	X or Y
female			
male			

(ii) Jon and Catrin have two daughters. Catrin is expecting their third child. State the probability that this child will be a boy. [1]

Probability =

3430U401

11

8

